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Homework 1

I included an extra page at the end of the pdf containing screenshots of my whole python file.

Show your work. Include any code snippets that you used to generate answers. Complete this assignment individually.

- 1) What are the two main types of attributes typically found in data.
[2 points]

Categorical and Quantitative

- 2) Consider the following matrix D; and answer all the questions. [14 points]

	X_1	X_2	X_3
x_1	1.2	3	1.7
x_2	2.4	5	2.4
x_3	4.8	35	1.2
x_4	6.6	60	3.1
x_5	-0.5	24	3.3
x_6	3.4	32	8.4
x_7	2.1	1	6.5

- a. What is the estimated mean of this data attribute X_1 ? [2 points]

$$\mu_1 = 2.857$$

```
C: > ALL_STORAGE_HERE > Classwork > Spring 2
1  import numpy as np
2
3  data = [[1.2, 3, 1.7],
4          [2.4, 5, 2.4],
5          [4.8, 35, 1.2],
6          [6.6, 60, 3.1],
7          [-0.5, 24, 3.3],
8          [3.4, 32, 8.4],
9          [2.1, 1, 6.5]]
10
11 matrix_array = np.array(data)
12
13 X0 = matrix_array[:, 0]
14 print(X0)
15
16 X0MEAN = X0.mean()
17 print(X0MEAN)
```

- b. What is the estimated covariance between attributes X_2 and X_3 ? [2 points]

```
#Prob 2b
X2 = matrix_array[:, 1]
X3 = matrix_array[:, 2]
covariance = np.cov(X2, X3)[0,1]
print(covariance)
```

-1.4

- c. What is the estimated multi-dimensional mean $\hat{\mu}$ of this data matrix? (Your answer should be a vector) [2 points]

```
#Prob 2c
X2mean = X2.mean()
X3mean = X3.mean()
print("-----PROBLEM 2c-----")
print("(")
print(X1mean)
print(X2mean)
print(X3mean)
print(")")
```

$\hat{\mu}$ (2.9 22.9 3.8)

- d. What is the estimated covariance $\hat{\sigma}_{12}$ of the attributes X_1 and X_2 ? [2 points]

```
#Prob 2d
cov_array = np.vstack((X1,X2))
cov_matrix = np.cov(cov_array)
print("-----PROBLEM 2d-----")
print(cov_matrix)
```

$\sigma_{12} = 36.893$

e. What is the covariance matrix of D ? [2 points]

```
#Prob 2e
Dcov_matrix = np.cov(data, rowvar=False)
print("-----PROBLEM 2e-----")
print(Dcov_matrix)

# Covariance of D
#[[ 5.47952381  36.89285714 -0.23666667]
# [36.89285714 467.14285714 -1.40000000]
# [-0.23666667 -1.40000000  7.05333333]]
```

f. What is the correlation coefficient between the attributes X_1 and X_2 ? [2 points]

```
#Prob 2f
X1X2cov = np.cov(X1, X2)[0, 1]
VarX1 = Dcov_matrix[0,0]
VarX2 = Dcov_matrix[1,1]
Ans2f = X1X2cov / (VarX1 * VarX2)
print("-----PROBLEM 2f-----")
print(Ans2f)
```

0.0144

g. What is the total variance of this matrix D ? [2 points]

```
#prob 2g
totalVariance = np.var(data, ddof=1)
print("-----PROBLEM 2g-----")
print(totalVariance)
```

233.0413

3) Consider the following 5-dimensional vectors: [6 points]

$$a = (1.2 \quad -2.3 \quad 4 \quad 7.1 \quad -3.12)$$

$$b = (23.2 \quad 3 \quad 1.2 \quad -3.21 \quad 5)$$

$$c = (8.2 \quad -4.6 \quad 2 \quad 1 \quad -2)$$

a. What is the $\|a - c\|_2$? [2 points]

```
#prob 3a
a = np.array([1.2, -2.3, 4, 7.1, -3.12])
b = np.array([23.2, 3, 1.2, -3.21, 5])
c = np.array([8.2, -4.6, 2, 1, -2])

L2_AC = np.linalg.norm(a - c)
print("-----PROBLEM 3a-----")
print(L2_AC)
```

9.836

b. What is the $\|b - a\|_1$? [2 points]

```
#prob 3b
L1_BA = np.linalg.norm(b - a, ord=1)
print("-----PROBLEM 3b-----")
print(L1_BA)
```

48.53

c. What is the angle between the vectors a and c ? [2 points]

```
#prob 3c
cosTheta = np.dot(a, c) / (np.linalg.norm(a) * np.linalg.norm(c))
angle = np.degrees(np.arccos(cosTheta))
print("-----PROBLEM 3c-----")
print(angle)
```

62.302°

4) Consider the following matrix D.

$$D = \begin{pmatrix} & X_1 & X_2 & X_3 \\ x_1 & 36.6 & Mild & 32 \\ x_2 & 38 & Severe & 21 \\ x_3 & 37 & Extreme & 67 \\ x_4 & 39 & Extreme & 11 \\ x_5 & 27 & Moderate & 71 \end{pmatrix}$$

- a. Use the One-Hot encoding method to transform the categorical data into numerical data in the following matrix. You can assume that the attribute X_2 can only contain 4 values: $\{Mild, Moderate, Severe, Extreme\}$ [2 points]

$$D = \begin{pmatrix} & X_1 & X_2 & X_3 \\ x_1 & 36.6 & 1 & 32 \\ x_2 & 38 & 3 & 21 \\ x_3 & 37 & 4 & 67 \\ x_4 & 39 & 4 & 11 \\ x_5 & 27 & 2 & 71 \end{pmatrix}$$

mild = 1
moderate = 2
Severe = 3
Extreme = 4

b. What is the value of $\|x_3 - x_5\|$ (based on the transformed data matrix)? [2 points]

```
#prob 4b
D = np.array([[36.6, 1, 32],
              [38, 3, 21],
              [37, 4, 67],
              [39, 4, 11],
              [27, 2, 71],
              ])

x3 = D[2]
x5 = D[4]

dist = np.linalg.norm(x3 - x5)
print("-----PROBLEM 4b-----")
print(dist)
```

10.954

5) The following questions reference the Heart Disease data set from the UCI Machine Learning Repository:

<https://archive.ics.uci.edu/ml/datasets/Heart+Disease>

Answer the following questions about the dataset.

a. How many rows (entities/instances) are there in this dataset? [1 point]

303 (in cleaveland data set)

b. How many attributes are in this dataset? [1 point]

14

c. What is kind of data is stored in the attribute "cigs"? [1 point]

quantitative

```

1 import numpy as np
2 import pandas as pd
3
4 data = [[1.2, 3, 1.7],
5         [2.4, 5, 2.4],
6         [4.8, 35, 1.2],
7         [6.6, 60, 3.1],
8         [-0.5, 24, 3.3],
9         [3.4, 32, 8.4],
10        [2.1, 1, 6.5],]
11
12 matrix_array = np.array(data)
13
14 #Prob 2a
15 X1 = matrix_array[:, 0]
16 X1mean = X1.mean()
17 print("-----PROBLEM 2a-----")
18 print(X1mean)
19
20 #Prob 2b
21 X2 = matrix_array[:, 1]
22 X3 = matrix_array[:, 2]
23 covariance = np.cov(X2, X3)[0,1]
24 print("-----PROBLEM 2b-----")
25 print(covariance)
26
27 #Prob 2c
28 X2mean = X2.mean()
29 X3mean = X3.mean()
30 print("-----PROBLEM 2c-----")
31 print("")
32 print(X1mean)
33 print(X2mean)
34 print(X3mean)
35 print("")
36
37 #Prob 2d
38 cov_array = np.vstack((X1,X2))
39 cov_matrix = np.cov(cov_array)
40 print("-----PROBLEM 2d-----")
41 print(cov_matrix)
42
43 #Prob 2e
44 Dcov_matrix = np.cov(data, rowvar=False)
45 print("-----PROBLEM 2e-----")
46 print(Dcov_matrix)
47
48 #Prob 2f
49 X1X2cov = np.cov(X1, X2)[0, 1]
50 VarX1 = Dcov_matrix[0,0]
51 VarX2 = Dcov_matrix[1,1]
52 Ans2f = X1X2cov / (VarX1 * VarX2)
53 print("-----PROBLEM 2f-----")
54 print(Ans2f)
55
56 #prob 2g
57 totalVariance = np.var(data, ddof=1)
58 print("-----PROBLEM 2g-----")
59 print(totalVariance)
60
61 #prob 3a
62 a = np.array([1.2, -2.3, 4, 7.1, -3.12])
63 b = np.array([23.2, 3, 1.2, -3.21, 5])
64 c = np.array([8.2, -4.6, 2, 1, -2])
65
66 L2_AC = np.linalg.norm(a - c)
67 print("-----PROBLEM 3a-----")
68 print(L2_AC)
69
70 #prob 3b
71 L1_BA = np.linalg.norm(b - a, ord=1)
72 print("-----PROBLEM 3b-----")
73 print(L1_BA)
74
75 #prob 3c
76 cosTheta = np.dot(a, c) / (np.linalg.norm(a) * np.linalg.norm(c))
77 angle = np.degrees(np.arccos(cosTheta))
78 print("-----PROBLEM 3c-----")
79 print(angle)
80
81 #prob 4b
82 D = np.array([[36.6, 1, 32],
83              [38, 3, 21],
84              [37, 4, 67],
85              [39, 4, 11],
86              [27, 2, 71],
87              ])
88
89 x3 = D[2]
90 x5 = D[48]
91
92 dist = np.linalg.norm(x3 - x5)
93 print("-----PROBLEM 4b-----")
94 print(dist)

```